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Models and Tools for Cyber-Physical Systems
Exercise sheet 6

Exercise 1: Worker Process

Consider the timed automaton for a *Worker* in Figure 1. Assuming that the Worker works for 70 minutes, what is the **minimum and maximum** number of hits (i.e., taking the transition labeled with *hit*) if the worker

(a) starts with a rest (i.e., starts in the state $(\texttt{Rest}, x = 0, y = 0)$), or

(b) is immediately ready to work (i.e., starts in the state $(\texttt{Work}, x = 0, y = 0)$).

Provide valid executions that prove your claims. You may summarize repeated transitions appropriately.

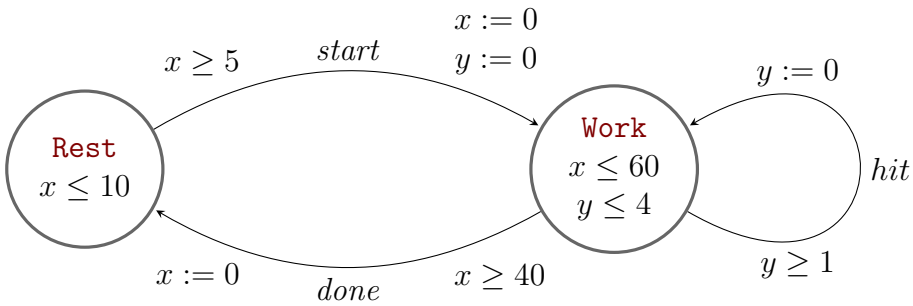

Figure 1: Timed automaton for the worker.

Exercise 2: Alarm Timer

An *Alarm Timer* is a timed process which can be set to time-out after a prescribed time period has elapsed. Provide a timed automaton model of an alarm timer T that can be set to time-out after 5, 10, and 30 minutes by receiving a *set* event and reading one of the input values 5, 10, and 30 on the input channel *in*. After the prescribed time period, T signals the time-out by the output action *to*. It is required that the alarm timer can be reset with a new time-out period at any given moment, in particular before the previously set time period has elapsed.

Exercise 3: Fischer's Protocol
Download the UPPAAL model of Fischer's Protocol (FischersProtocol.xml) from Moodle and examine it using the simulator. In addition use verification to check whether the following properties are satisfied or not[1]:

- Does the protocol satisfies the mutual exclusion property:

$$\forall\Box\neg(\texttt{Process}(1).\texttt{Crit} \wedge \texttt{Process}(2).\texttt{Crit})$$

- Does the protocol satisfies the non-starvation property: if a process wants to enter its critical section then it eventually does

$$\texttt{Process}(1).\texttt{Test} \rightarrow \texttt{Process}(1).\texttt{Crit}$$

- Is the protocol deadlock-free, i.e.

$$\forall\Box\neg\text{deadlock}$$

Perform verification of the above properties for varying values of the constants `D1` and `D2`. In case the properties does not hold, try to obtain a counter-example. Also, perform the verifications for varying numbers of processes.

Exercise 4: Worker Process – revisited
Let us revisit the *Worker* process from Exercise 1. Model the Worker in UPPAAL and provide queries to prove your claims of the minimum and maximum hits.

Exercise 5: Symbolic Exploration of Timed Automata
Consider the timed automaton $\mathcal{A}$ in Figure 2 with two clocks `x` and `y`. Perform a symbolic zone-base reachability analysis of $\mathcal{A}$. Which of the locations `D`, `E`, and `F` are reachable? Describe the reachable zones upon entry and after delay for the locations `A`, `B`, and `C`.

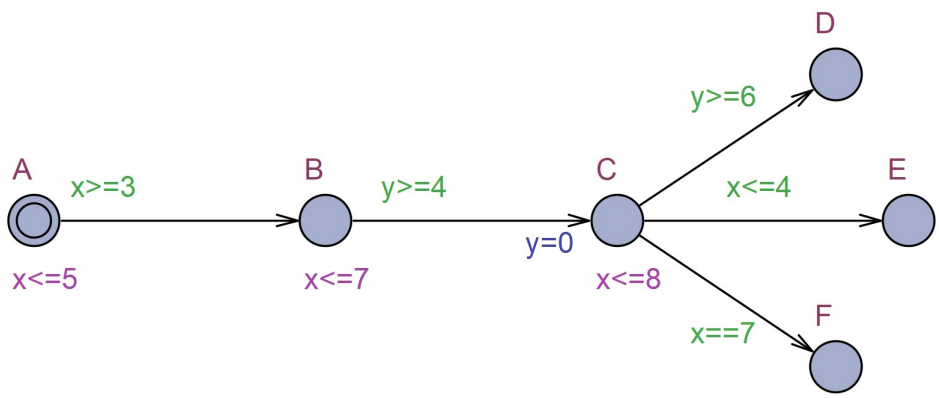

Figure 2: Timed automaton $\mathcal{A}$.

[1]You can find UPPAAL's documentation on queries here: `https://docs.uppaal.org/language-reference/query-syntax/symbolic_queries/`.